Lecture 10: Model Evaluation

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Model Evaluation

- Model needs to be evaluated. Therefore, we need some model evaluation techniques

- Confusion Matrix
  - A confusion matrix shows that number of correct and incorrect predictions made by the classification model compared to actual outcomes (target value) in the data
  - The matrix is N×N, where N is the number of classes in the target variable

- Cancer Classification Model
  - Train Logistic Regression model
  - y=1 for cancer, y=0 otherwise

Confusion Matrix

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Terminologies

- Accuracy- the proportion of total number of prediction that were correct
- Positive predicted value or Precision- the proportion of positive cases that are correctly identified
- Negative predicted value- the proportion of negative cases that are correctly identified
- Sensitivity or Recall- the proportion of actual positive cases that are correctly identified
- Specificity- the proportion of actual negative cases that are correctly identified
- F1 Score- Harmonic mean of precision and recall

Precision

- Positive predicted value or Precision- the proportion of positive cases that are correctly identified

- Precision

= TP/Total predicted positive

=TP/TP+FP

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Negative predicted value

- Negative predicted value - the proportion of negative cases that are correctly identified

- Negative predicted value

= TN/Total predicted Negative

=TN/FN+TN

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Sensitivity or Recall

- Sensitivity or Recall- the proportion of actual positive cases that are correctly identified

- Recall

= TP/Total Actual Positive

=TP/TP+FN

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Specificity

- Specificity- the proportion of actual negative cases that are correctly identified

- Specificity

= TN/Total Actual Negative

=TN/FP+TN

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Accuracy

- Accuracy- the proportion of total number of prediction that were correct

- Accuracy

= Total trueprection

/Total predicted value

=TP+TN/TP+FP+FN+TN

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

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Confusion Matrix		Target (Actual)			
		Positive	Negative		
Model (Predicted)	Positive	a (TP)	b (FP)	Positive Predicted value	a/(a+b)
	Negative	c (FN)	d (TN)	Negative Predicted Value	d/(c+d)
		Sensitivity	Specificity	Accuracy = (a+d)/(a+b+c+d)	
		a/(a+c)	d/(b+d)		

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Confusion Matrix		Target (Actual)			
		Positive	Negative		
Model (Predicted)	Positive	a (TP)	b (FP)	Positive Predicted value	a/(a+b)
	Negative	c (FN)	d (TN)	Negative Predicted Value	d/(c+d)
		Sensitivity	Specificity	Accuracy	
		a/(a+c)	d/(b+d)	(a+d)/(a+b+c+d)	

Trading of Precision and Recall

- Logistic Regression: $0 \leq h_\theta(x) \leq 1$
  - Predicted 1 if, $h_\theta(x) \geq 0.5$
  - Predicted 0 if, $h_\theta(x) < 0.5$

Trading of Precision and Recall

- Suppose, we want to predict y=1 (cancer) only if we are very confident

- Logistic Regression: $0 \le h_\theta(x) \le 1$
  - Predicted 1 if, $h_\theta(x) \ge 0.7$
  - Predicted 0 if, $h_\theta(x) < 0.7$

- Precision =TP/TP+FP
- Recall=TP/TP+FN
- Precision will be higher and Recall will be lower

Trading of Precision and Recall

- Suppose, we want to avoid missing too many cases of cancer, y=1 or we want to avoid too many false negative

- Logistic Regression: $0 \leq h_\theta(x) \leq 1$
  - Predicted 1 if, $h_\theta(x) \geq 0.3$
  - Predicted 0 if, $h_\theta(x) < 0.3$

- Precision =TP/TP+FP
- Recall=TP/TP+FN
- Recall will be higher and Precision will be lower

Trading of Precision and Recall

- For model, its possible to get different precision and recall value
- same model for different thresholds, its possible to get different precision and recall value

- Precision and Recall are very important for unbalanced / skewed dataset

- How to compare precision and recall value?

F1 Score

- F1 Score= 2 *(Precision*Recall) / Precision+Recall

- If Precision=0 and Recall=0, F1 Score=0
- If Precision=1 and Recall=1, F1 Score=1

- F1 Score is between 0 to 1

Average vs F1Score

	Precision	Recall	Average	F1 Score
Algorithm 1	0.5	0.4	0.45	0.44
Algorithm 2	0.7	0.1	0.4	0.175
Algorithm 3	0.02	1.0	0.51	0.039

- Average can be same for different precision and recall value
- F1 gives a more distinct value

Receiver Operator Characteristic (ROC)

- If we make the threshold very low,
  - Predicted 1 if, $h_\theta(x) \geq 0.1$
  - Predicted 0 if, $h_\theta(x) < 0.1$

- We will predict too many output as cancer (y=1)

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	TP HIGH	FP HIGH
	Negative	FN LOW	TN LOW

ROC

- If we make the threshold very low,
  - Predicted 1 if, $h_\theta(x) \geq 0.1$
  - Predicted 0 if, $h_\theta(x) < 0.1$
- We will predict too many output as cancer (y=1)

True Positive rate/Recall is high

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	TP HIGH	FP HIGH
	Negative	FN LOW	TN LOW

ROC

- If we make the threshold very low,
  - Predicted 1 if, $h_\theta(x) \geq 0.1$
  - Predicted 0 if, $h_\theta(x) < 0.1$
- We will predict too many output as cancer (y=1)

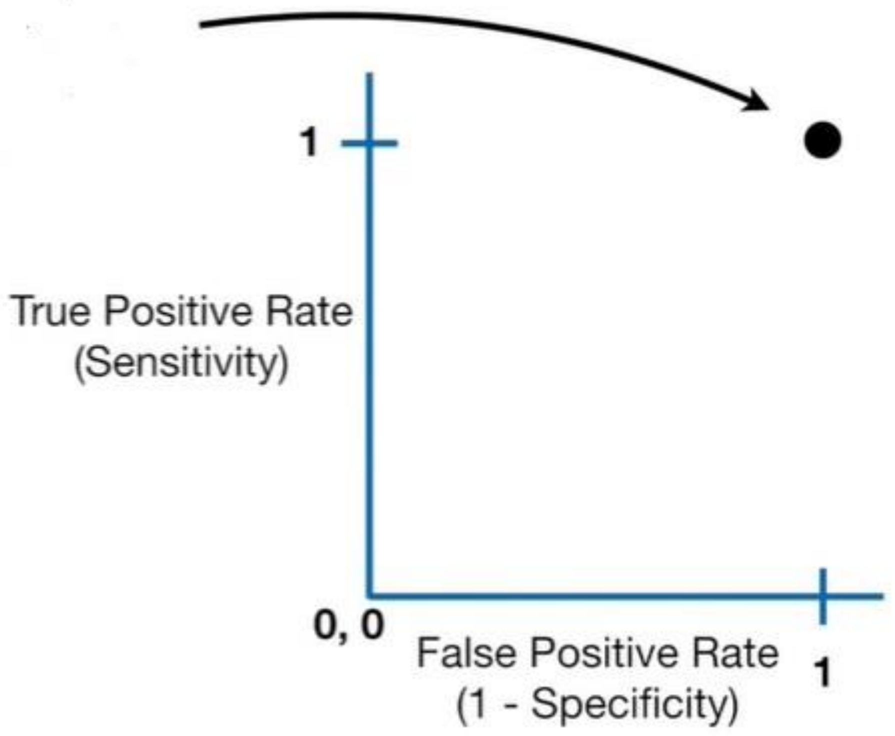
•True Positive rate/Recall is high

•False Positive rate is also high

Confusion Matrix		Target (Actual)	
		Positive	Negative
Model (Predicted)	Positive	TP HIGH	FP HIGH
	Negative	FN LOW	TN LOW

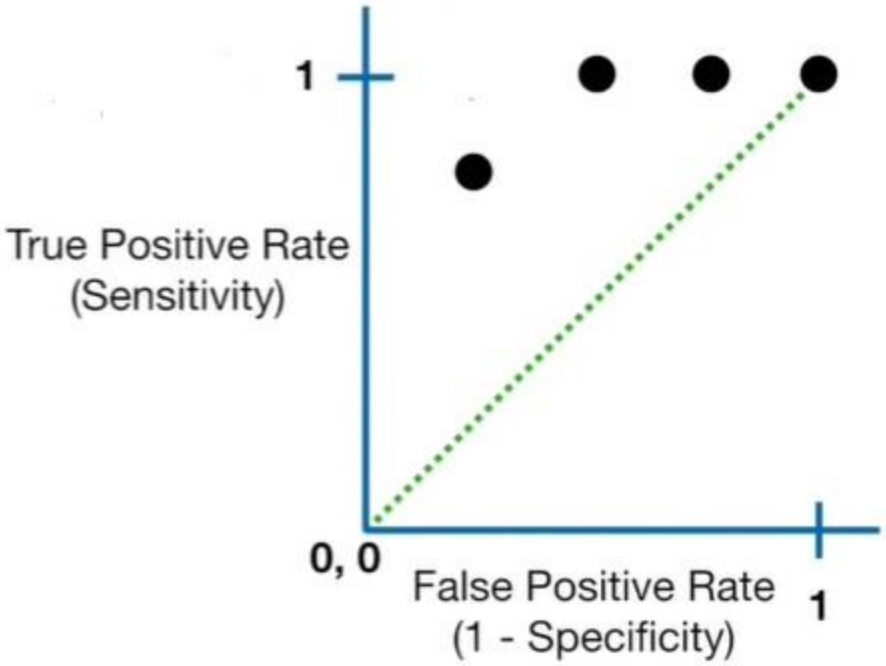
ROC

- Point (1,1) means even though we have correctly classified all of the Cancer sample, we have incorrectly classified all the samples that are not Cancer



ROC

- If we increased the threshold, The point will keep moving



ROC

- If we make the threshold very High,
  - Predicted 1 if, $h_\theta(x) \geq 0.99$
  - Predicted 0 if, $h_\theta(x) < 0.99$
- We will predict zero output as cancer (y=1)

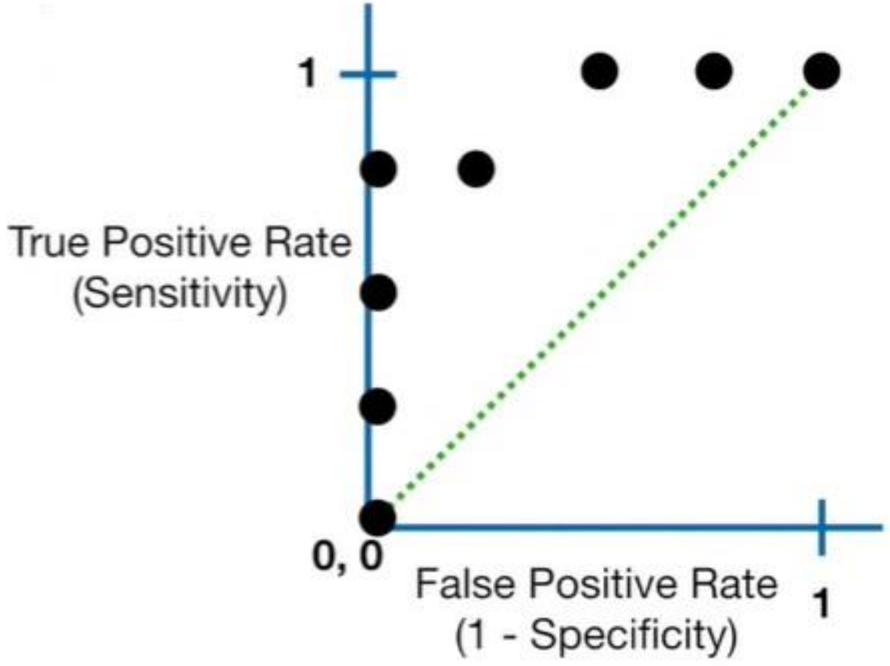
•True Positive rate/Recall is LOW

•False Positive rate is also LOW

Confusion Matrix		Target (Actual) Positive	Target (Actual) Negative
Model (Predicted)	Positive	TP LOW	FP LOW
	Negative	FN HIGH	TN HIGH

ROC

- If we put the threshold very high=1, both true positive rate and false positive rate will be 0.

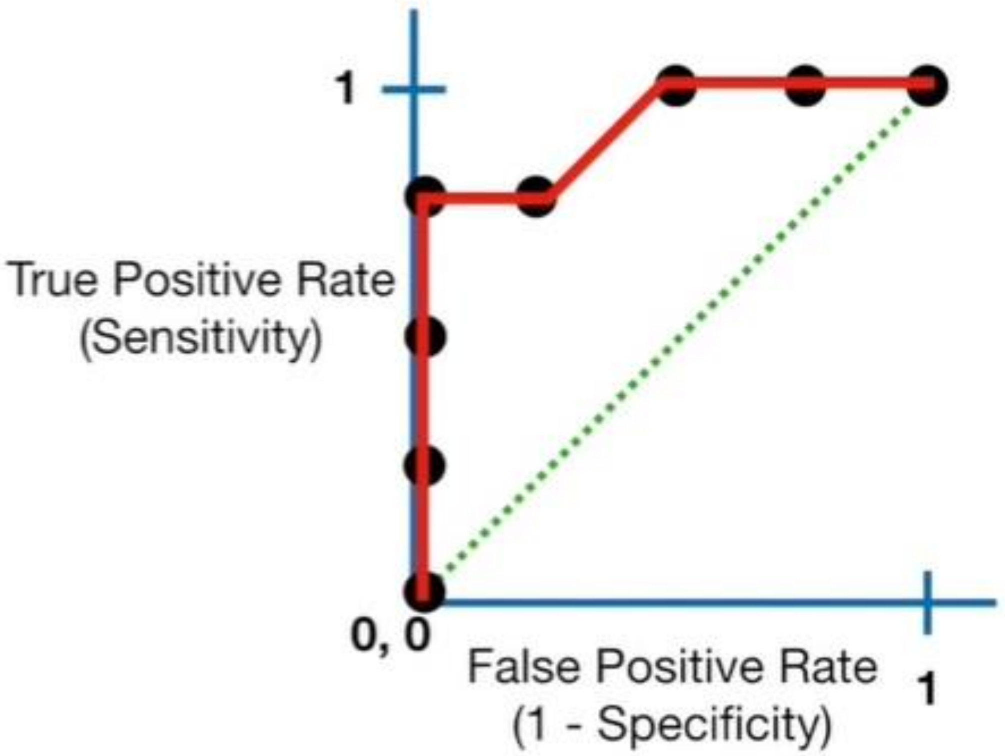


ROC

- ROC means **Receiver Operator Characteristic**. It summarizes all of the information in a simple way

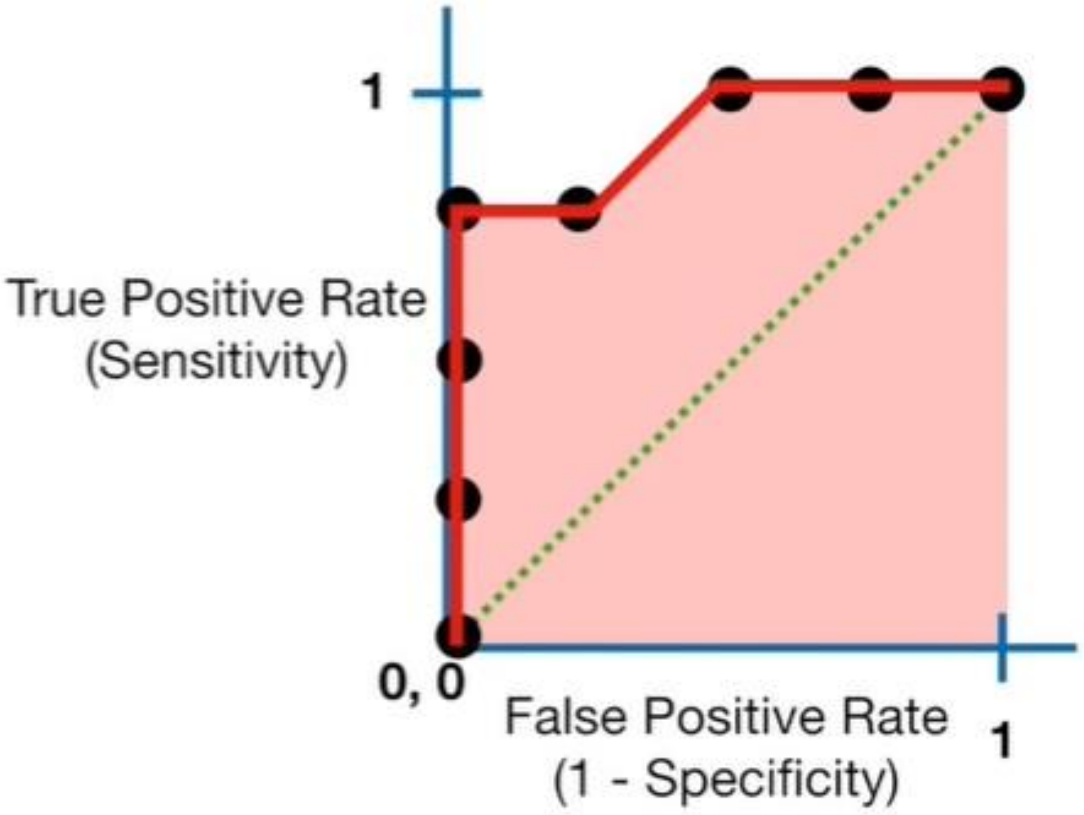
- ROC Curve summarizes all the confusion matrices that each threshold produces

- ROC curve: TPR vs FPR



AUC

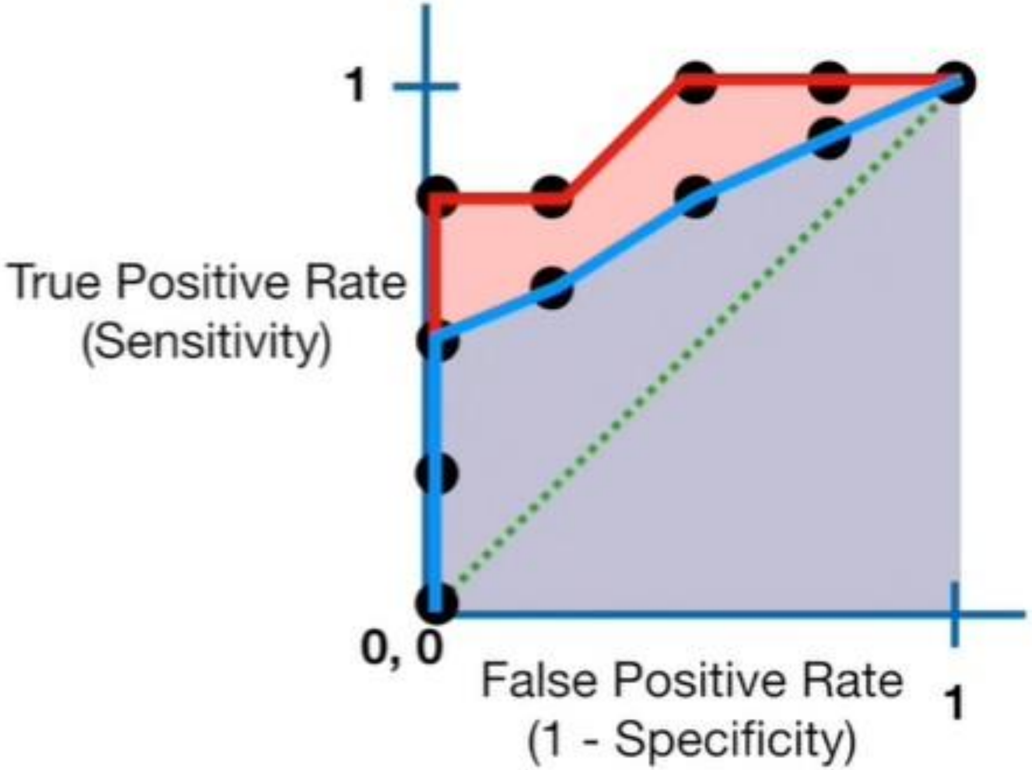
- AUC: Area under the curve



AUC

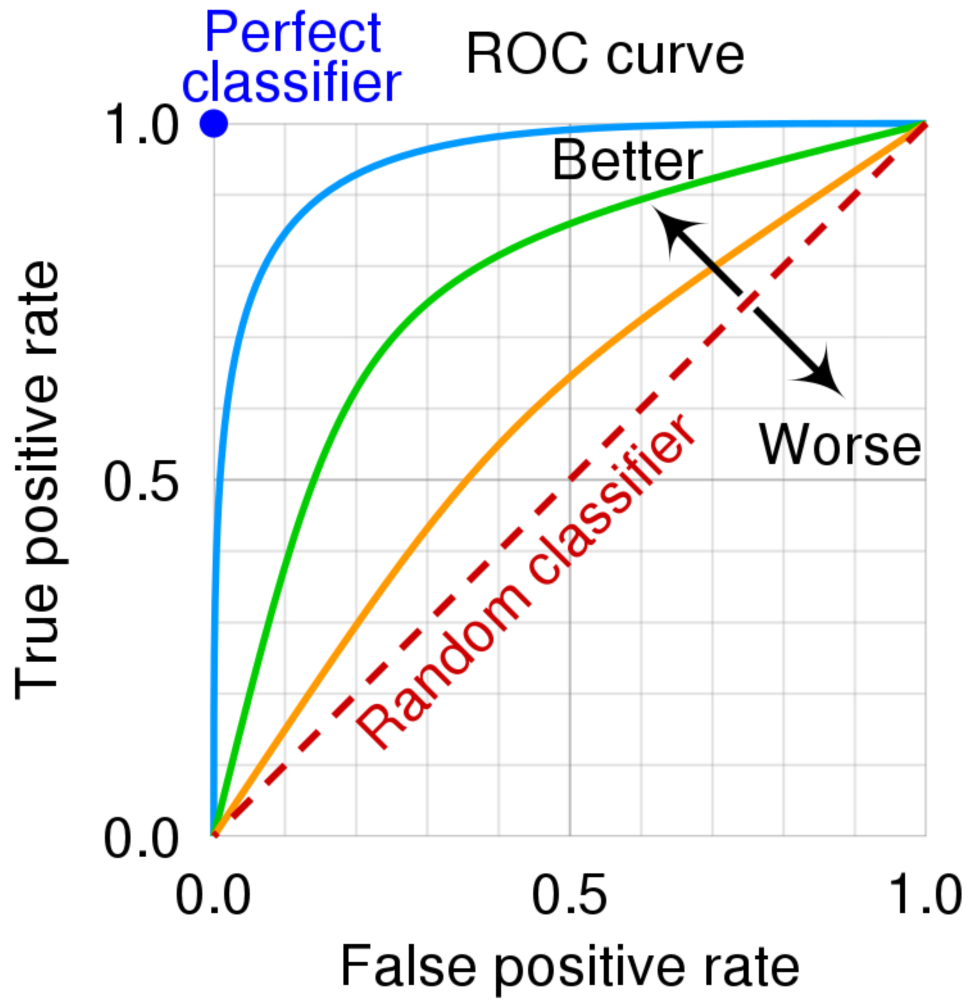
- The AUC curve make it easy to compare different models ROC curve

- The greater the AUC curve is, the better the model is performing



ROC and AUC

- The closer AUC and ROC curve to 1, the better model is performing



Confusion Matrix for Multiclass Problem

- For Three classes

Confusion Matrix		Target (Actual)		
		Class 1	**Class 2**	**Class 3**
Model (Predicted)	**Class 1**	TP		
	Class 2		TP	
	Class 3			TP

Confusion Matrix for Multiclass Problem

- For Three classes

Confusion Matrix		Target (Actual)		
		Dog	**Cat**	**Bird**
Model (Predicted)	**Dog**	TP		
	Cat		TP	
	Bird			TP

Confusion Matrix for Multiclass Problem

- For Dog

Confusion Matrix		Target (Actual)		
		Dog	**Cat**	**Bird**
Model (Predicted)	**Dog**	TP	FP	FP
	Cat	FN	TN	TN
	Bird	FN	TN	TN

Confusion Matrix for Multiclass Problem

- For Cat

Confusion Matrix		Target (Actual)		
		Dog	**Cat**	**Bird**
Model (Predicted)	**Dog**	TN	FN	TN
	Cat	FP	TP	FP
	Bird	TN	FN	TN

Confusion Matrix for Multiclass Problem

- For Bird

Confusion Matrix		Target (Actual)		
		Dog	**Cat**	**Bird**
Model (Predicted)	**Dog**	TN	TN	FN
	Cat	TN	TN	FN
	Bird	FP	FP	TP